Data Structures & Algorithms

1. Sorting (20 marks)

Present the idea of the Insertion Sort algorithm for array. For a given random data, compare and explain the runtime efficiency of Insertion Sort with that of Bubble Sort and Selection Sort.

2. Sorting (15 marks)

Given the sequences S:

17, 72, 99, 32, 58, 70, 44, 12, 23

Provide the detailed steps to arrange S in descending order using the Insertion Sort algorithm, showing the changes in S at each iteration.

3. Linked list (35 marks)

Given the sorted (ascending order) SinglyLinkList class below:

- a) (15 marks) Complete the insertNode method that inserts a Node with a given integer into the sorted list at the right location.
- b) (20 marks) Complete the deleteNode method that deletes all occurrences of a given integer value from the list.

```java
public class SinglyLinkList {
    private Node first;
    public void insertNode(int value){
    // YOUR CODE
    }
    public void deleteAllNodes(int value) {
    // YOUR CODE
    }
```

```
}

public class Node {
    private int key;
    private Node next;
}
```

4. Queue and Stack (20 marks)

- a. (10 marks) Show the queue after each operation of the following sequence of MAX PRIORITY QUEUE operations, if executed on an initially empty queue?
  enqueue(1), enqueue(20), dequeue(), enqueue(30), enqueue(4), dequeue(), dequeue(), enqueue(50), enqueue(0), dequeue(), enqueue(2), enqueue(6), dequeue(), dequeue(), enqueue(100), dequeue(), dequeue()
- b. (10 marks) Show the stack after each operation of the following series of stack operations, if executed upon an initially empty stack?
  push(10), push(20), pop(), push(0), push(1), pop(), pop(), push(2), push(30), pop(), push(3), push(4), pop(), pop(), push(100), pop(), pop()

5. Queue Reversal (10 marks)

Given the sequences S:
Given an integer k and a queue of integers.
Implement a function to reverse the order of the first k elements of the queue while keeping the other elements in the same original order.

Only following operations are allowed on the queue:

- enqueue(x): Add an item x to rear of queue
- dequeue(): Remove an item from the front of the queue
- size(): Returns the number of elements in the queue.
- front(): Finds front item.

Example

- Input: q = 1 2 3 4 5, k = 3

- Output: 3 2 1 4 5
- Explanation: After reversing the first 3 elements from the given queue the resultant queue will be 3 2 1 4 5.
- Input: q = 4 3 2 1, k = 4
- Output: 1 2 3 4
- Explanation: After reversing the first 4 elements from the given queue the resultant queue will be 1 2 3 4.